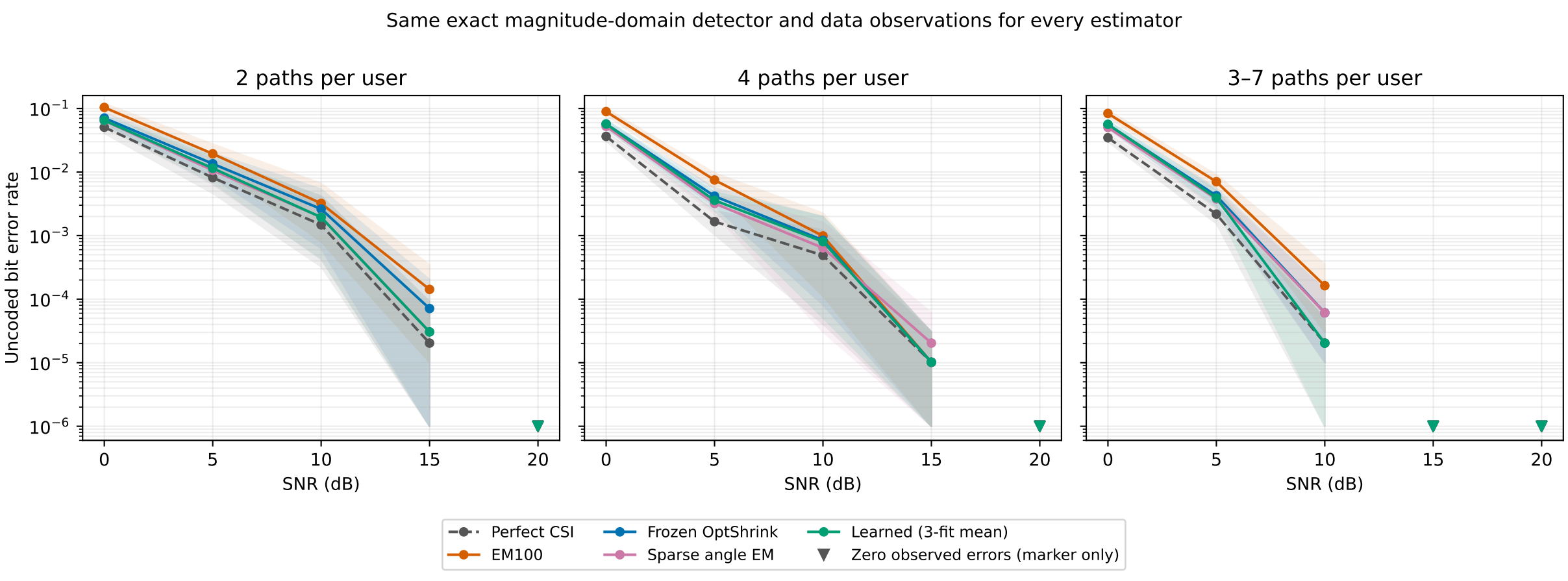
Same exact magnitude-domain detector and data observations for every estimator
2 paths per user
4 paths per user
3–7 paths per user
Uncoded bit error rate
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
0
5
10
15
20
SNR (dB)
Perfect CSI
EM100
Frozen OptShrink
Sparse angle EM
Learned (3-fit mean)
Zero observed errors (marker only)
Shading: 95% channel-block bootstrap intervals. Zero-error markers at 10^{-6} are a display choice, not a BER estimate or bound.